



When Should Language Models Search?

A Topology-Aware Analysis of Reasoning Strategies

M.Tech Thesis Defense

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Gap:

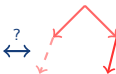
- CoT = single linear trace; brittle on multi-hop
- Search helps *sometimes* — but when?
- Trace-based routers: expensive, misled by fluency

CoT



Single path

A*

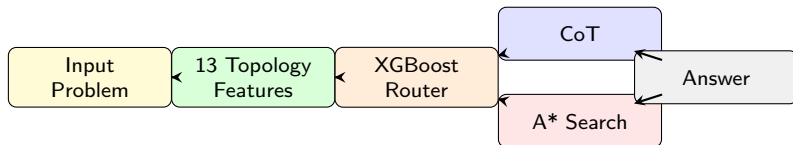


Multiple paths

Which one to use? → Our router decides

Aim:

- Predict CoT vs A* from **input structure alone**
- 13 pre-generation topology features
- 8 benchmarks $\times$ 3 scales (QA + Math + Code)



- **A* search:** Priority queue over reasoning traces ($f = g + h$), heuristic-guided
- **13 features:** Extracted from input text only (connectives, hops, branching coefficient)
- **No LLM call needed** for routing — only text analysis
- Router trained via 5-fold stratified CV per dataset



A* over Reasoning Traces:

- State = partial trace; expand via LLM step generation
- $g(n) = d_n + \sum(1-c_t) \quad h(n) = \omega_d(H_{\min}-d_n) + \omega_c(1-E(n))$
- Admissible heuristic (proven); budget-limited search

13 Pre-Generation Topology Features:

- | | |
|-----------------------|-------------------------|
| 1 Question length | 8 Context sentences |
| 2 Context length | 9 Named entities |
| 3 Logical connectives | 10 Linearity score |
| 4 Negation markers | 11 Answer options |
| 5 Hop indicators | 12 Depth estimate |
| 6 Comparison words | 13 Branching complexity |
| 7 Question marks | |

$$BC = \frac{\text{ctx_sents} \times \text{hop}}{\text{q_len}}$$

Predicts A* advantage: $R^2 = 0.94$

Results I: Main Results



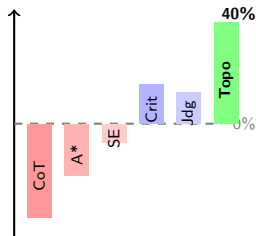
Dataset	8B		0.5B		14B	
	CoT	A*	CoT	A*	CoT	A*
StrategyQA	64.5	74.8	53.6	53.4	76.0	75.6
LogicQA	45.8	44.9	24.0	20.2	57.9	58.1
HotpotQA	76.8	80.1	8.4	30.0	60.2	83.3
GSM8K	85.0	86.1	26.8	14.5	61.6	80.6
MATH	78.0	79.1	32.8	16.2	44.0	47.2
HumanEval	67.5	64.4	33.7	20.2	77.9	76.1
MBPP	39.8	36.8	24.0	20.8	46.6	45.0
CodeContests	77.4	87.0	49.8	68.2	99.6	94.4

- **Neither dominates.** A* wins on multi-hop/algorithmic; CoT wins on linear tasks
- A* **+19pp** on GSM8K (14B); **+18.4pp** on CodeContests (0.5B)
- Oracle gap up to **14pp** → routing is worthwhile

Results II: Router Comparison



Router	Gap%
CoT only	-24.6
A* only	-10.9
Sem. Entropy	-3.9
LLM-Critic	+9.6
LLM-Judge	+6.6
Random Forest	-3.3
Topology	+40.0



- **Fluency–Correctness Asymmetry:** 38% of fluent traces are *wrong*
- Post-hoc judges conflate readability with correctness
- Topology router sidesteps this — routes by **structure alone**



Wei, J. et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS*.



Yao, S. et al. (2024). Tree of thoughts: Deliberate problem solving with LLMs. *NeurIPS*.



Hao, S. et al. (2023). Reasoning with language model is planning with world model. *EMNLP*.



Snell, C. et al. (2024). Scaling LLM test-time compute optimally. *arXiv:2408.03314*.



Wang, X. et al. (2023). Self-consistency improves chain of thought reasoning. *ICLR*.



Hart, P. et al. (1968). A formal basis for heuristic determination of minimum cost paths. *IEEE Trans*.



Chen, M. et al. (2021). Evaluating large language models trained on code. *arXiv:2107.03374*.



Kossen, J. et al. (2024). Semantic entropy probes. *arXiv:2406.15927*.



Thank You